
What would be the speed of a satellite that orbits 500 km above the earth?

$$a_C = \frac{v^2}{r} \quad \Sigma F_C = ma_C = \frac{mv^2}{r} \quad v = \frac{2\pi r}{T} \quad F_G = \frac{Gm_1m_2}{r^2}$$

$$G = 6.67 \times 10^{-11} \text{ Nm}^2/\text{kg}^2$$

$$\Sigma F = ma$$

$$w = mg$$

$$f = \mu N$$

Earth:	Mass	$5.98 \times 10^{24} \text{ kg}$
	Radius (mean)	$6.38 \times 10^3 \text{ km}$
Moon:	Mass	$7.35 \times 10^{22} \text{ kg}$
	Radius (mean)	$1.74 \times 10^3 \text{ km}$
Sun:	Mass	$1.99 \times 10^{30} \text{ kg}$
	Radius (mean)	$6.96 \times 10^5 \text{ km}$
Earth-Sun Distance (mean)		$1.496 \times 10^8 \text{ km}$
Earth-Moon Distance (mean)		$3.84 \times 10^5 \text{ km}$

What would be the speed of a satellite that orbits 500 km above the earth?

$$a_C = \frac{v^2}{r} \quad \Sigma F_C = ma_C = \frac{mv^2}{r} \quad v = \frac{2\pi r}{T} \quad F_G = \frac{Gm_1m_2}{r^2}$$

$$G = 6.67 \times 10^{-11} \text{ Nm}^2/\text{kg}^2$$

$$\Sigma F = ma$$

$$w = mg$$

$$f = \mu N$$

Earth:	Mass	$5.98 \times 10^{24} \text{ kg}$
	Radius (mean)	$6.38 \times 10^3 \text{ km}$
Moon:	Mass	$7.35 \times 10^{22} \text{ kg}$
	Radius (mean)	$1.74 \times 10^3 \text{ km}$
Sun:	Mass	$1.99 \times 10^{30} \text{ kg}$
	Radius (mean)	$6.96 \times 10^5 \text{ km}$
Earth-Sun Distance (mean)		$1.496 \times 10^8 \text{ km}$
Earth-Moon Distance (mean)		$3.84 \times 10^5 \text{ km}$
